

The Coefficient of the a_0 – Λ Relation: a Zero-Mode Theorem for Local MOND Actions, Two Failed Repairs, a Four-Form Reframing, and an H_0 Degeneracy

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Abstract

The empirical relation $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with $\kappa \simeq \frac{1}{2}$ (measured 0.465 ± 0.076 from the baryonic Tully–Fisher intercept and 0.551 ± 0.043 by a distance-free estimator) has never been derived. We ask whether the covariant candidate action of the de Sitter–MOND closure programme — a khronometric clock, a dynamical MOND scalar with a bounded-boost kernel, and a coherence operator — can relate a_0 to Λ at all, and we answer in the negative with a theorem and three computed dead ends. (i) The MOND primitive $J(Y)$ enters the static field equations only through J' and the FLRW background only through $\Lambda + (2 - K_B)J(0)/2 + K(Q_0)/2$; with Λ explicit, no equation of the action relates the two scales (a normalisation zero mode). (ii) Deleting Λ and fixing the primitive’s zero by an empty Newtonian vacuum yields a *negative* vacuum energy of 0.4–0.7% of ρ_Λ ; the constant $4\pi^4/15$ of the RAR primitive that would give $\kappa \approx 1$ belongs to the unsaturated branch that the bounded-boost theorem forbids. (iii) A sequestering-type global constraint tying Λ to the spacetime average of the scalar’s Lagrangian misses by five orders of magnitude today and tends to zero in the de Sitter future. (iv) Promoting a_0 to a conserved four-form flux, $a_0 = \beta\sqrt{G}|q|$, reverses the sign, makes $a_0 \propto \sqrt{G\rho_\Lambda}$ structural with the flux amplitude cancelling, and leaves the coefficient as the coupling ratio $Z/\beta^2 = 7.96$; the induced environmental a_0 is invisible in galaxies, switches the scalar off above $155 a_0$, and shifts the Gaia DR4 wide-binary boost at 2 kAU by -0.019 , about one sigma. On the observable side the only coefficient-free alternative left standing, the horizon form $a_0 = c^2/(2\pi L_{\text{dS}})$ with $\kappa = 0.461$, is separated from $\frac{1}{2}$ by 8.5%, below the 9.5% mass-budget floor of the baryonic Tully–Fisher route, and is exactly degenerate with the H_0 tension: $\kappa = \frac{1}{2}$ on the Planck H_0 and $\kappa = 0.461$ on the SH0ES H_0 predict the same a_0 to 0.2%. We conclude that $\kappa = \frac{1}{2}$ is an empirical boundary condition that this class of local actions cannot derive, and that its determination is a precision problem gated by the stellar mass-to-light zero point, the absolute gas scale and H_0 . Every number is produced by a committed script with checks that can fail.

1 Two claims, kept separate

Writing $\rho_\Lambda = \Lambda c^2/(8\pi G)$, the relation $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ is $a_0 = c^2\sqrt{\Lambda/32\pi}$, or in natural units $a_0 = M_\Lambda^2/(2M_P)$ with $M_\Lambda^4 = \rho_\Lambda$ and $M_P = G^{-1/2}$: a seesaw with an exact factor of two. Two statements must be kept apart. The *empirical* one, $\kappa_{\text{obs}} \simeq \frac{1}{2}$, is a measurement with a 10–16% error and is the frozen target of the programme. The *theoretical* one, that some completion forces $4a_0^2 = Gc^2\rho_\Lambda$ with no adjustable coefficient, has never been established, and the rule adopted here is that no mechanism counts unless a_0 and Λ begin independent and the equations remove one degree of freedom: no

32 π placed in a coupling, no vacuum constant chosen after integrating, no change of convention. We also record that the data do not single out the rational $\frac{1}{2}$: the horizon form $c^2/(2\pi L_{\text{dS}})$ gives $\kappa = \sqrt{8\pi/3}/2\pi = 0.461$, and four simple candidates sit inside two sigma of the measurements.

2 The action and the zero-mode theorem

The candidate action, in the convention of the closure programme, is

$$S = \int d^4x \sqrt{-g} \frac{1}{16\pi G} \left[R - 2\Lambda - c_1 T_1 - c_2 T_2 - c_3 T_3 + c_4 T_4 + 2(2 - K_B) J^\mu \partial_\mu \phi - K(Q) - (2 - K_B) J(Y) \right] + S_m, \quad (1)$$

with $n_\mu = -\partial_\mu \tau / N$ the clock's unit normal, $J^\mu = n^\nu \nabla_\nu n^\mu$ its acceleration, $Q = n \cdot \partial \phi$, $Y = q^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$ the scalar's spatial gradient squared, $K = K_2(Q - Q_0)^2$, $c_1 = -c_3 = K_B$, and the coherence operator $\xi^2 |\nabla_\perp V|^2$ carried inside J 's argument. The kernel is a single function $J_Y(s) = s/\Delta(s)$, where $g_\phi = a_0 \Delta(s)$ is the scalar's boost at Newtonian acceleration $g_N = s a_0$; the carried kernel is ν_{RAR} , $\Delta(s) = s/(e^{\sqrt{s}} - 1)$, saturated at its maximum $s_{\text{sat}} = 2.540$, $\Delta_{\text{sat}} = 0.6476$ (the bounded-boost theorem forbids the non-monotone branch beyond it). The primitive in units of a_0^2 is $j(s) = J/a_0^2 = 2 \int_0^s s' d\Delta$, with $j(s_{\text{sat}}) = I = 0.4525$ (0.0719 for the superseded exponential carrier).

Theorem 1 (zero mode) *The static field equations of the action contain J only through J' , so $J \rightarrow J + C$ is a symmetry of the local sector; on the FLRW background ($Y = 0$, $Q = Q_0$) the constant enters only as $\Lambda_{\text{eff}} = \Lambda + (2 - K_B)J(0)/2 + K(Q_0)/2$. With Λ explicit and free, no equation of the action relates a_0 to Λ .*

The proof is a two-line variation and is executed symbolically (script **k01**, checks K1–K2): the scalar equation is $\nabla \cdot (J_Y \nabla \phi) = \nabla^2 \Psi$ for a generic J , and the constant part of the minisuperspace Lagrangian is $-a^3[2\Lambda + (2 - K_B)J(0) + K(Q_0)]$. This is OpenAI's "outcome 3" for the action as written, and it is not a computation but an inspection: an interpolation law fixes derivatives of the primitive while ρ_Λ reads its additive zero, and those are independent pieces of information.

3 Repair 1: the Λ -free vacuum

Delete the explicit Λ and fix the primitive's zero by the only principle available to a local action, an empty Newtonian vacuum: $J = 0$ at the saturated end. The background then carries $\rho_{\text{vac}} = (2 - K_B)J(0)/(16\pi G) = -(2 - K_B)I a_0^2/(16\pi G)$. Three facts follow (**k01**, K3–K6).

Sign. $I = 2 \int s d\Delta > 0$ for every kernel of the class, because the scalar's gradient term carries the attractive sign and its primitive rises toward the Newtonian end. The vacuum energy is negative for both kernels, both $K_B \in \{0, 0.25\}$ and both footings.

Size. $|\rho_{\text{vac}}|/\rho_\Lambda = (2 - K_B)I\kappa^2/16\pi = 0.0045$ (ν_{RAR} , canonical) to 0.0007 (carrier); the needed $|J(0)|$ is $16\pi a_0^2/((2 - K_B)\kappa^2) = 100 a_0^2$ against a primitive whose whole span is $0.45 a_0^2$, a factor 220.

The $4\pi^4/15$. The RAR primitive's celebrated constant is what the unsaturated, non-monotone branch returns: $2 \int_0^\infty s d\Delta = -4\pi^4/15$ for ν_{RAR} and -2 for the carrier. Read as the vacuum energy anyway it predicts $\kappa = \sqrt{60/((2 - K_B)\pi^3)} = 0.98\text{--}1.05$, at least 6.8σ from the data; the coefficient that would put it at $\frac{1}{2}$ is $16\pi/(\kappa^2 \cdot 4\pi^4/15) = 7.7$ against the action's $2 - K_B \leq 2$, and $(2 - K_B)^2$ gives 0.70–0.80, also out. No term of the action supplies a 7.7.

4 Repair 2: a global constraint

The one class of principle that removes an additive zero mode without choosing a constant is a global constraint of the vacuum-energy-sequestering type: Λ becomes a Lagrange multiplier fixed by the spacetime average of the field Lagrangians. The only sector with scale a_0^2 is the MOND scalar, whose on-shell, curl-free Lagrangian density is pointwise $(2 - K_B)a_0^2 F(s)/(16\pi G)$ with $F = 2s\Delta - j$ ($\rightarrow \frac{4}{3}s^{3/2}$ in deep MOND, linear beyond saturation). Averaged over the cosmic peculiar-acceleration field (Maxwellian, rms 0.005–0.012 a_0 today from $v_{\text{rms}} = 300\text{--}600 \text{ km s}^{-1}$, scaling as $D(a)/a^2$), the ratio $\langle L_\phi \rangle / \rho_\Lambda$ is $0.6\text{--}2.3 \times 10^{-5}$ today, is 10^{-8} of that when the average is extended to $10 t_0$ because the de Sitter future dominates the volume, and would reach unity only if the universe sat at $s_* = 54\text{--}89 a_0$ on average (k02, G2–G4). A halo term at a hundred times its real filling factor changes nothing. Closed on magnitude, independently of sign and of the $O(1)$ convention.

5 Reframing: the four-form promotion

A three-form gauge potential with four-form field strength $F = q\epsilon$ carries positive vacuum energy without propagating modes, and for a vacuum action $P(q)$ the gravitating energy is the Legendre form $\varepsilon = qP_q - P$ rather than $-P$, because q depends on the metric through the volume element. Promote the MOND scale to the flux, $a_0 = \beta\sqrt{G}|q|$, with $P = Zq^2/2$. The primitive’s constant becomes $b\beta^2 q^2$ with $b = (2 - K_B)I/16\pi$, its gravitating energy is $+ba_0^2/G$, and the sign of Repair 1 is reversed without touching the attractive gradient term (k04, F1; an independent symbolic audit of the same reduced model passed twelve identities). Both a_0 and ρ_Λ are now set by one conserved flux, so $a_0 \propto \sqrt{G\rho_\Lambda}$ is structural and the amplitude cancels:

$$\kappa^2 = \frac{a_0^2}{G\varepsilon_{\text{vac}}} = \frac{2\beta^2}{Z + 2b\beta^2}, \quad \kappa = \frac{1}{2} \iff Z/\beta^2 = 8 - 2b = 7.96. \quad (2)$$

The coefficient has become a ratio of two couplings that nothing fixes: “why 32π ” is now “why $Z = 8\beta^2$ ”, a cleaner and still open question.

The promotion has a consequence that must be checked rather than inherited. Since J depends on a_0 , the four-form equation $\partial_\mu(\partial\mathcal{L}/\partial q) = 0$ reads $q[Z + (2 - K_B)\beta^2(s\Delta - j)/8\pi] = Zq_0$ with $\partial J/\partial a_0|_Y = (2/a_0)(J - YJ_Y)$ and $YJ_Y = g_\phi g_N$. In the saturated regime the equation is linear in q : $a_0^{\text{loc}} = a_0(1 - g_N/g_*)$ with $g_* = 8\pi Z a_0 / ((2 - K_B)\beta^2 \Delta_{\text{sat}}) = 155 a_0 = 1.4 \times 10^{-8} \text{ m s}^{-2}$, and above g_* the only solution is $q = 0$, a kink of $|q|$: the scalar switches off. The environmental a_0 is 1.2% low at the RAR knee and 64% low at $100 a_0$, yet the radial acceleration relation moves by less than 0.002 dex because the boost is already a small fraction of g_N there (F3). Inside 205 AU of the Sun the scalar is off, so the planetary monopole residual vanishes without the coherence length; the Cassini quadrupole is set at $s \approx 1$ where $a_0^{\text{loc}} = 0.997 a_0$, so the coherence length is still required (F4). For a $1.5 M_\odot$ wide binary a_0 is 15% low at 2 kAU, a shift $\delta\gamma_v = -0.019$ (canonical) / -0.015 (alt) in the innermost Gaia DR4 bin, about one sigma at the frozen error model (F5); the effective flux stiffness stays positive (F6). This variant is recorded, not registered, in Amendment 11 of the DR4 pre-registration.

6 The observable side: $\frac{1}{2}$ against 0.461, and H_0

Repairs 1 and 2 exhaust the principles that act on the scalar’s zero; a positive Λ must come from the geometric or boundary sector, where the only coefficient-free identification in hand is $a_0 = c^2/(2\pi L_{\text{dS}})$,

i.e. $\kappa = \sqrt{8\pi/3}/2\pi = 0.4607$. (This is not Milgrom’s 1999 de Sitter–Unruh construction, which fixes $a_0 = 2cH_\Lambda$ and is excluded at 15.6σ in this programme.) Script `k03` puts it against what the observable side can do:

- the two coefficients differ by 8.5% in a_0 (0.036 dex); a 3σ separation needs 2.8% on a_0 , whereas the baryonic Tully–Fisher route carries a 9.47% mass-budget floor ($f_* + f_{\text{gas}} = 1$ forces the stellar mass-to-light and gas calibrations to trade), a factor 3.3 short and a floor rather than a statistics limit;
- Gaia DR4 gives 21% on a_0 ($d \ln \gamma_v / d \ln a_0 = 0.1155$) and cannot contribute;
- the BTFR value 0.465 ± 0.076 is 0.5σ from $\frac{1}{2}$ and 0.1σ from 0.461; the distance-free value 0.551 ± 0.043 is 1.2σ and 2.1σ ; combined, $\ln[L(\frac{1}{2})/L(0.461)] = +1.4$, undecided;
- at fixed Ω_Λ the prediction scales as κH_0 , so $\kappa = \frac{1}{2}$ on the Planck $H_0 = 67.4$ and $\kappa = 0.461$ on the SH0ES $H_0 = 73.0$ predict the same a_0 to 0.2%: the coefficient question is numerically the H_0 tension. Holding $\Omega_m h^2$ to the CMB instead puts the SH0ES reading at 0.447, 3% from the horizon form.

Run the other way, a frozen $\kappa = \frac{1}{2}$ turns a_0 into an H_0 meter: the distance-free a_0 gives $H_0 = 74 \pm 6$ and the BTFR a_0 gives $63 \pm 10 \text{ km s}^{-1} \text{ Mpc}^{-1}$, straddling the tension. The measured κ values also sit on a mixed scale, Planck’s ρ_Λ against SPARC’s $H_0 = 73$ Hubble-flow distances, and a_0 from the BTFR scales as the inverse square of distance; the distance-free estimator avoids this and is the one to trust.

7 What a derivation must supply, and what we conclude

A derivation of the half within a local action must (a) fix the absolute zero of the scalar’s primitive by a principle, (b) supply a positive sign, which the scalar sector cannot, and (c) produce $\rho_\Lambda c^2 = 4a_0^2/G$, an $O(1)$ multiple of a_0^2/G with no 16π , which places it in the geometric or boundary sector. The four-form extension achieves (b) and turns the form into structure at the price of one free coupling ratio; nothing achieves (a) and (c) together. We therefore hold $\kappa = \frac{1}{2}$ as an empirical boundary condition of the programme, with 0.461 recorded as the one un-excluded principle-shaped alternative, and we state that the coefficient will be settled by a stellar mass-to-light zero point, an absolute gas scale and an H_0 that has stopped moving, not by more theory of the present class. The dark-sector completion hunt of the programme is frozen behind this result; its most recent candidate, a thermal relic, was closed the same day as a pincer with no interior between N_{eff} , the radial acceleration relation and the cluster profile.

Reproducibility

All numbers come from four scripts in `kappa_closure/` of the public repository (`k01_zero_mode_theorem_and_lambda_free_vacuum.py`, `k02_global_constraint_average.py`, `k03_half_vs_two_pi_precision.py`, `k04_four_form_promotion_consistency.py`; commits `d5cba7d34`, `28c61fe68`, `2fb80ca12`), each with checks that can fail and whose FAIL lines mark requirements the framework does not meet: 3 of 6, 3 of 4, 4 of 4 and 2 of 6 respectively. Both a_0 footings (9.3619×10^{-11} and $1.1279 \times 10^{-10} \text{ m s}^{-2}$) are carried throughout. The wide-binary

registration is Amendment 11 of `prep_2026/gaia_dr4_prep/PREREGISTRATION_DR4.md` (commit `5a865128f`, append-only, hash-stamped). The zero-mode argument and the four-form construction were proposed in an exchange with an OpenAI model on 6 September 2026 and are audited, not adopted, here.